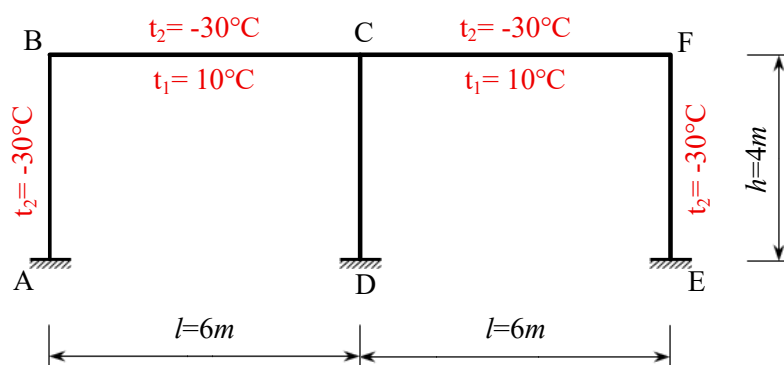
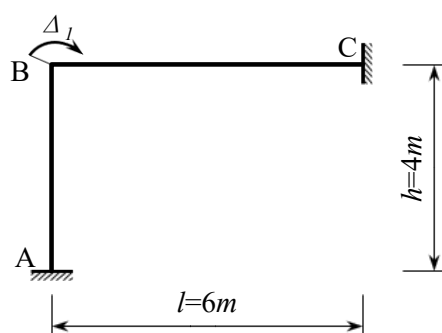


做下图示结构 M 图，钢架各杆为矩形截面，截面高为 0.6m，各杆 EI 相同。



解(1)由于改结构对称故取一半进行分析

取 ABC 进行分析如图:



共有一个基本未知量: θ_B

(2)杆端弯矩

固端弯矩查表求得

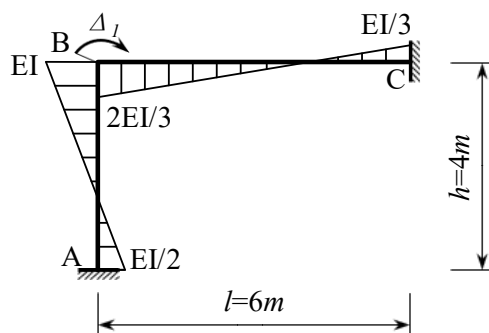
$$M_{BA} = 4i = \frac{4EI}{l} = EI$$

$$M_{AB} = 2i = \frac{2EI}{l} = \frac{EI}{2}$$

$$M_{BC} = 4i = \frac{4EI}{l} = \frac{2}{3}EI$$

$$M_{CB} = 2i = \frac{2EI}{l} = \frac{EI}{3}$$

如图 $\bar{M}$ 所示



B 处弯矩平衡得

$$k_{11} = EI + \frac{2}{3}EI$$

$$k_{11} = \frac{5}{3}EI$$

(3)温度作用下的杆端弯矩:

温度导致弯曲形变产生的杆端弯矩

$$M_{BA} = M_{AB} = \frac{\alpha EI \Delta t}{h} = \frac{200}{3} \alpha EI$$

$$M_{BC} = M_{CB} = \frac{\alpha EI \Delta t}{h} = \frac{200}{3} \alpha EI$$

B 处弯矩平衡得

$$F_{1t}(M) = 0$$

温度导致的纵向的形变量

$$\Delta_{BA} = \alpha t_0 l = -40\alpha$$

$$\Delta_{BC} = \alpha t_0 l = -60\alpha$$

温度导致的轴向变形产生的弯矩

$$M_{BA} = M_{AB} = \Delta_{BC} \frac{6i}{l} = \frac{45}{2} \alpha EI$$

$$M_{BC} = M_{CB} = \Delta_{BA} \frac{6i}{l} = \frac{20}{3} \alpha EI$$

B 处弯矩平衡

$$F_{1t}(N) = \frac{95}{6} \alpha EI$$

求解 Δ_1 :

$$k_{11}\Delta_1 + F_{1t}(M) - F_{1t}(N) = 0$$

$$k_{11}\Delta_1 = F_{1t}(N)$$

$$\Delta_1 = \frac{19}{2} \alpha$$

根据所求量求解各个点处的弯矩

对柱

$$M_{\perp} = EI * \frac{19}{2} \alpha + \frac{200}{3} \alpha EI - \frac{45}{2} \alpha EI = \frac{161}{3} \alpha EI$$

$$M_{\overline{\gamma}} = \frac{EI}{2} * \frac{19}{2} \alpha - \frac{200}{3} \alpha EI + \frac{20}{3} \alpha EI = -\frac{221}{4} \alpha EI$$

对梁

$$M_{\text{左}} = \frac{2EI}{3} * \frac{19}{2} \alpha - \frac{200}{3} \alpha EI + \frac{20}{3} \alpha EI = -\frac{161}{3} \alpha EI$$

$$M_{\text{右}} = \frac{EI}{3} * \frac{19}{2} \alpha + \frac{200}{3} \alpha EI + \frac{20}{3} \alpha EI = \frac{153}{2} \alpha EI$$

最终弯矩图如下:

